

OBJECTIVE:

- To study and comprehend the internal circuit configurations of different logic families, focusing particularly on Diode Logic (DL) and Diode-Transistor Logic (DTL) gates.
- To investigate the functionality of a NOT gate constructed using transistors.
- To implement a combinational logic function using basic gates (AND, OR, NOT) at the gate level, based on a provided truth table.

INTRODUCTION:

A logic family is a classification of digital circuits that share a specific circuit design methodology, maintaining consistent voltage levels and power supply specifications. These circuits are fundamentally made up of logic gates, which in turn are built from basic electronic components like resistors, diodes, and transistors. Depending on the design and components used, logic families can be divided into several categories, such as:

- Diode Logic (DL)
- Diode-Transistor Logic (DTL)
- Resistor-Transistor Logic (RTL)
- Transistor-Transistor Logic (TTL)
- Integrated Injection Logic (IIL), and others.

LIST OF EQUIPMENT:

- DC Power Supply
- Breadboard
- Resistors
- Diodes
- Transistors (BJTs)
- Multimeter
- Connecting Wires

CIRCUIT DIAGRAM FOR AND GATE:

TRUTH TABLE:

(repeat for AND, OR, NAND, NOR, NOT)

JUSTIFICATION / EXPLANATION OF THE EXPERIMENTAL DATA:

1. AND Gate (Diode Logic – DL)

Concept:

In DL, an AND gate is designed using diodes in such a way that the output only goes HIGH when **all inputs** are HIGH. If any input is LOW, the corresponding diode conducts and pulls the output LOW.

Experimental Result:

- Output stayed HIGH only when **both inputs were HIGH**.
- In all other combinations, the diodes became active and forced the output to LOW.

2. OR Gate (Diode Logic – DL)

Concept:

In this setup, the diodes are arranged so that if **any input** is HIGH, it allows current to flow to the output node, resulting in a HIGH output. If all inputs are LOW, the output remains LOW.

Experimental Result:

- The output turned HIGH when **at least one input** was HIGH.
- Output was LOW **only when all inputs were LOW**.

3. NOT Gate

Concept:

A NOT gate, or inverter, can be realized using a transistor. When the input is HIGH, the transistor switches ON and pulls the output LOW. When the input is LOW, the transistor stays OFF, and the output is pulled HIGH through a resistor.

Experimental Result:

- Input HIGH → Output LOW
- Input LOW → Output HIGH

The output clearly demonstrated inversion of the input logic.

4. NAND Gate (DTL)

Concept:

A NAND gate combines an AND gate followed by a NOT gate. Diodes handle the AND function, and a transistor inverts the result. The output only becomes LOW if **all inputs** are HIGH.

Experimental Result:

- Output remained HIGH in all cases **except when both inputs were HIGH**, where it turned LOW.

5. NOR Gate (DTL)

Concept:

This circuit operates as an OR gate using diodes, followed by an inverter implemented using a transistor. The output is HIGH only when **all inputs** are LOW.

Experimental Result:

- Output was HIGH only when both inputs were LOW.
- For all other input combinations, the output was LOW.

ANSWERS TO REPORT QUESTIONS:

1. Analyze all the circuits with the help of truth table for positive logic

In positive logic systems, a logic '1' corresponds to a HIGH voltage (usually +5V), and logic '0' to LOW (0V). Here are the respective truth tables:

(a) AND Gate (Fig. 1a – Diode Logic)

Input A	Input B	Output
0	0	0
0	1	0
1	0	0

1	1	1
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Explanation:

Only when both inputs are HIGH does the output become HIGH. Any LOW input causes a diode to conduct, grounding the output.

(b) OR Gate (Fig. 1b – Diode Logic)

Input A	Input B	Output
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0	0	0
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0	1	1
---	---	---

1	0	1
---	---	---

1	1	1
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Explanation:

Any HIGH input makes the output HIGH by allowing current through the diode. Both inputs must be LOW for the output to be LOW.

(c) NAND Gate (Fig. 1c – DTL)

Input A	Input B	Output
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0	0	1
---	---	---

0	1	1
---	---	---

1	0	1
---	---	---

1	1	0
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Explanation:

This gate outputs LOW only when both inputs are HIGH due to the AND-then-invert configuration.

(d) NOR Gate (Fig. 1d – DTL)

Input A	Input B	Output
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0	0	1
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0	1	0
---	---	---

1	0	0
---	---	---

1	1	0
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Explanation:

The circuit inverts the result of an OR gate. So, the output is HIGH only when both inputs are LOW.

(e) NOT Gate (Fig. 1e – Transistorized)

Input	Output
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0	1
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1	0
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Explanation:

The transistor turns ON when input is HIGH, grounding the output. When input is LOW, the transistor stays OFF, and the output is pulled HIGH by the resistor.

2. Why two diodes have been used in the circuit connection of Fig. 1(d)? Explain.

Answer:

In Fig. 1(d), which implements a NOR gate using DTL, the two diodes are responsible for performing the OR function. Each diode connects one input to the output node. If either input is HIGH, its corresponding diode conducts, causing the output to be HIGH. This signal is then inverted by the transistor to yield the NOR behavior — output LOW when any input is HIGH. Thus, the use of two diodes enables proper OR logic before inversion.

3. What is the minimum voltage required to make the diodes forward biased in Fig. 1(a), Fig. 1(b), Fig. 1(c), and Fig. 1(d)?

Answer:

For standard silicon diodes, the typical forward voltage threshold is approximately 0.7V. This means the input voltage must be at least 0.7V to turn the diode ON (i.e., make it conduct current).

- In Fig. 1(a) (AND Gate),
- Fig. 1(b) (OR Gate),
- Fig. 1(c) (NAND Gate), and
- Fig. 1(d) (NOR Gate),

diodes are used as the core logic elements. In all these circuits, the minimum input voltage required to forward bias the diodes is 0.7V.

However, in practical digital circuits—especially using positive logic—a voltage level of approximately +5V is typically applied as logic HIGH. This ensures reliable operation and clear distinction between logic HIGH and LOW levels.

4. Describe the operation of the circuit diagram in Fig. 1(e)

Answer:

Fig. 1(e) demonstrates a NOT gate built using a BJT in common emitter mode. Here's how it operates:

- **When the input is LOW (0V):** The transistor remains OFF, so the collector is not conducting. The output node is then pulled HIGH by a connected resistor.
→ Input 0 results in Output 1.

- **When the input is HIGH (+5V):** The transistor turns ON and goes into saturation. It conducts heavily, creating a path to ground and pulling the output LOW.
→ Input 1 results in Output 0.

This setup ensures the circuit inverts the input logic, as expected from a NOT gate.

DISCUSSION:

In this experiment, we explored the construction and behavior of basic logic gates—AND, OR, NOT, NAND, and NOR—using discrete components, specifically focusing on Diode Logic (DL) and Diode-Transistor Logic (DTL) families. The gates were built on a breadboard using diodes, resistors, transistors (BJTs), and powered by a DC supply.

Through testing each gate against its truth table under **positive logic** (where +5V = logic 1 and 0V = logic 0), we confirmed that:

- **AND and OR gates** functioned as expected using diode-based configurations.
- **NOT gate** used a transistor to invert the input signal.
- **NAND and NOR gates** combined diode logic with transistor inversion, showcasing DTL behavior.

We also observed that the **diodes required a minimum of 0.7V** to conduct (forward bias), but +5V was used to ensure proper logic levels. The experimental outcomes matched theoretical expectations, reinforcing our understanding of how fundamental digital circuits operate using simple electronic components.

Overall, this lab effectively demonstrated the physical implementation of logic operations and highlighted the role of each component in digital logic design.